

Jul 28, 2022

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

Define the notion of hash function and its main properties: (a) Ease of computation, (b) Compression, (c) One-way, (d) Weak collision resistance, and (e) Strong collision resistance.

Question 2 [10 points]

Define the notions of Confidentiality, Integrity, and Availability.

Part 2

Question 3 [40 points]

Explain the notion of Multi-level security by illustrating it with the Bell-La Padula model; in particular discuss the notions of (a) security level, (b) need-to-know, (c) dominance relation, (d) no-read-up rule, (e) no-write-down rule.

Question 4 [10 points]

Briefly describe the notion of Data Protection Impact Assessment (DPIA) according to the Art. 35 of the GDPR.

Please, remember...

On each sheet, make sure to write:

Name

Surname

Student ID number

Question number